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Autoencoders

Motivation

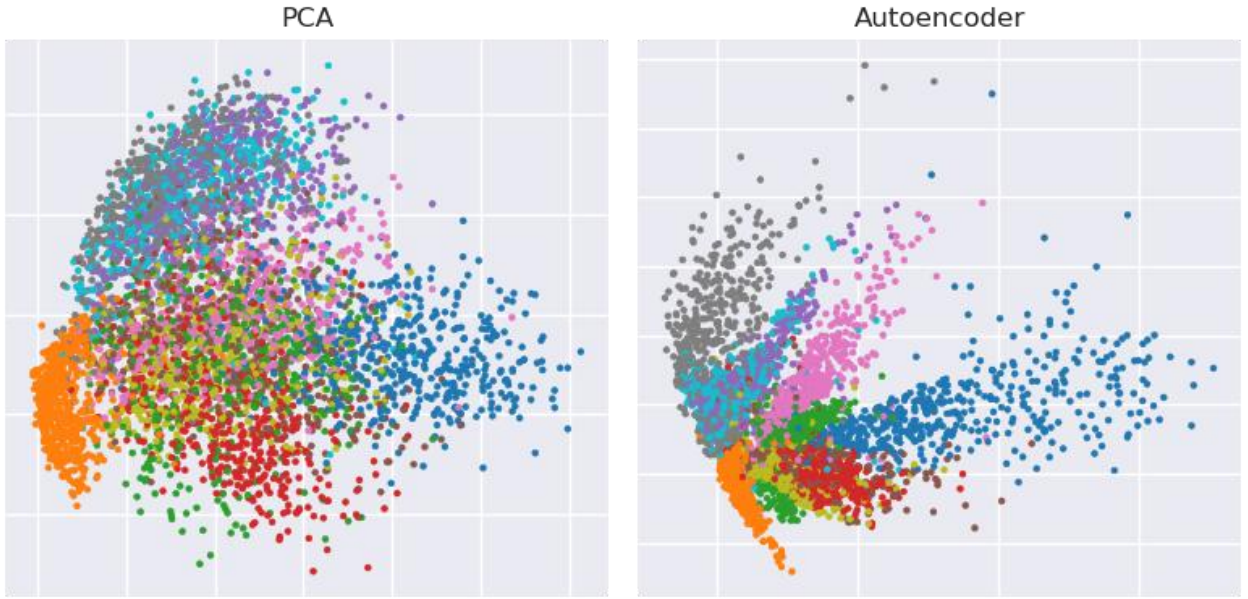
- No labels — so what do we use as the training target?
- The data itself. Reconstruct the input; learn a **representation** along the way.
- Day 3 flashback: PCA compressed data linearly.
- An autoencoder is the nonlinear version, learned by a neural network.

PCA vs AE

- Linear encoder/decoder + MSE → finds the **same subspace as PCA**

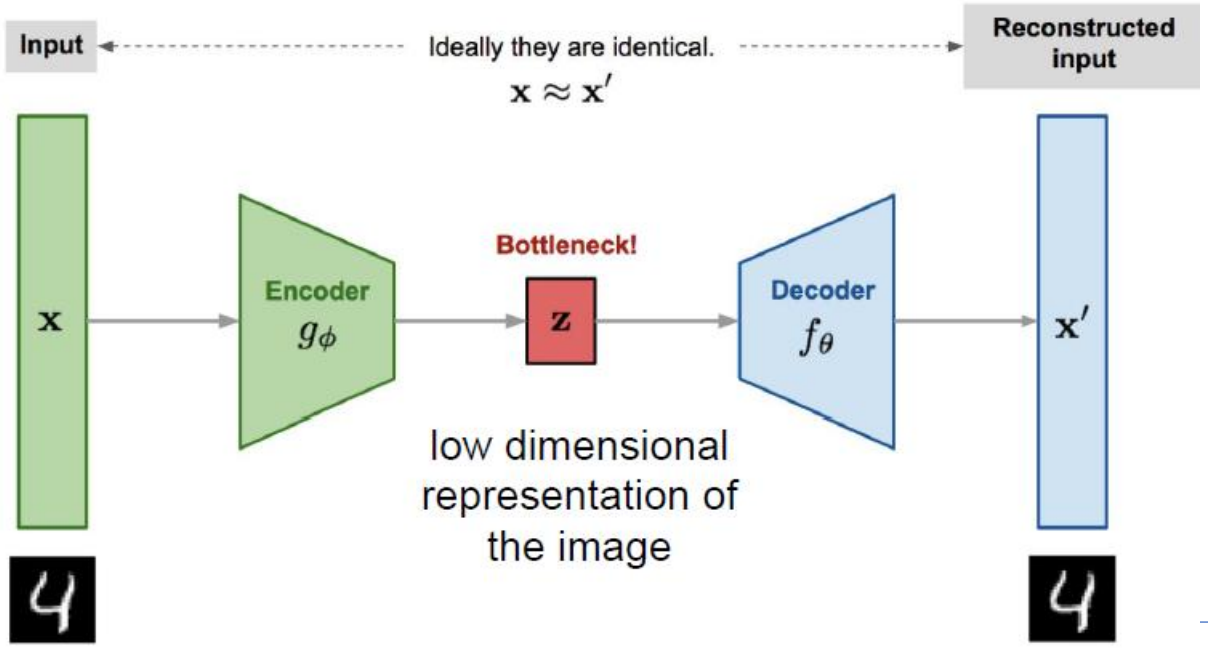
  → Actually…PCA is a linear AE

- Nonlinear activations → can flatten **curved** data structure PCA can't



Components

- **Encoder:** Performs dimensionality reduction and creates a latent space.
- **Decoder:** Reconstructs the input using the latent space.
- The **bottleneck** forces the network to keep only what matters
- Nothing new — just an MLP + backprop.

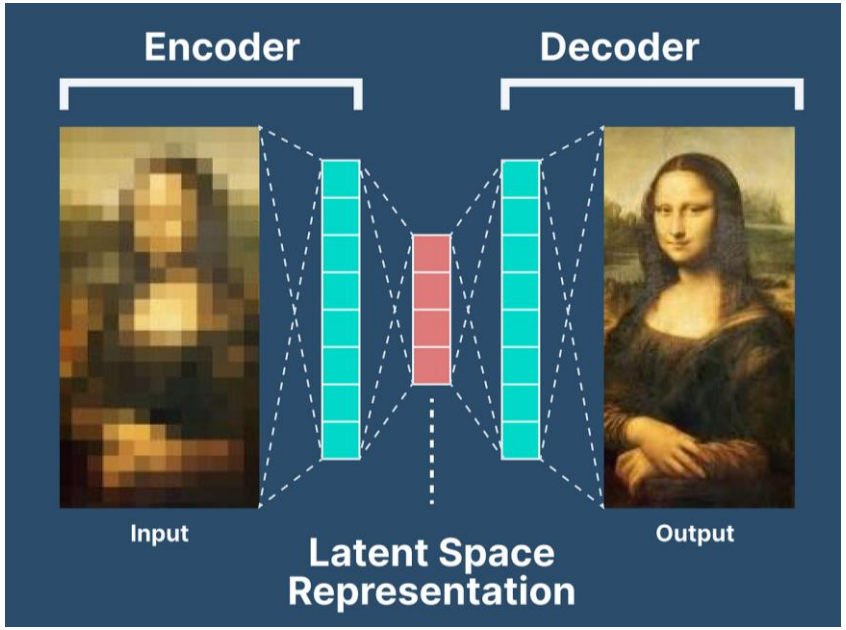


Applications

- Feature Extraction / Dimensionality Reduction
- Image Denoising
- Anomaly Detection

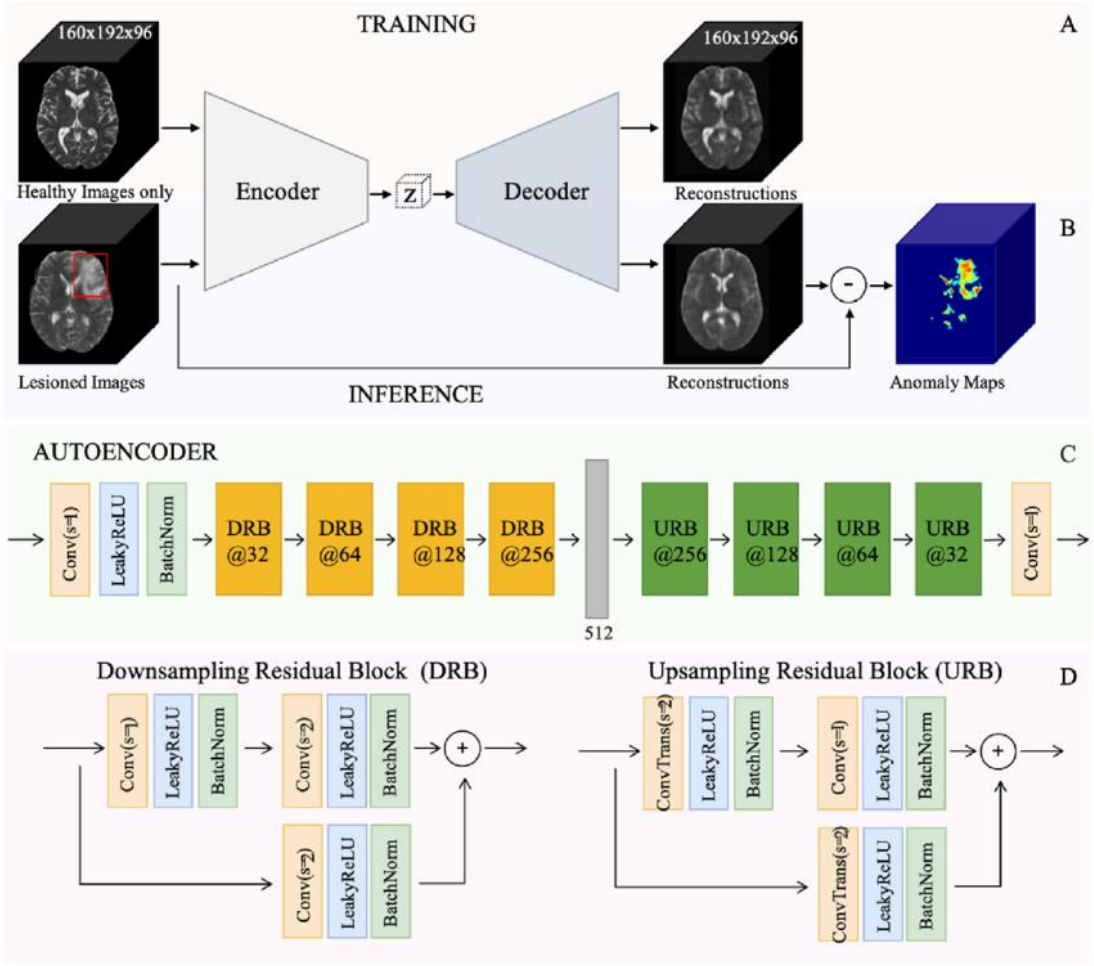
Application – Denoising

- Train: corrupted input → clean target
- Noise is unpredictable — the network's best strategy is to drop it and output the underlying structure
- Only the underlying signal survives reconstruction



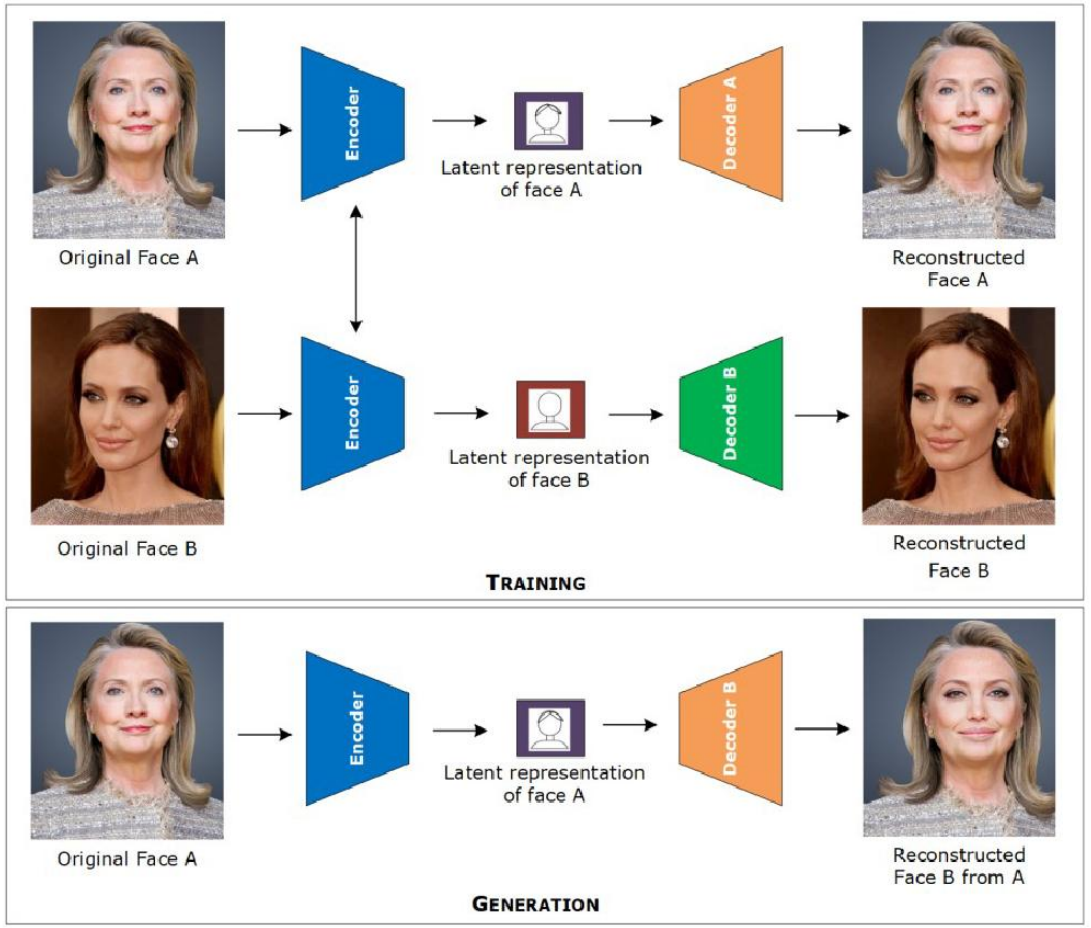
Application – Anomaly Detection

- Train on normal data only
- Anomalies reconstruct poorly → high reconstruction error
- Flag anything with error above a threshold
- Used in: manufacturing defect detection, ECG monitoring, fraud



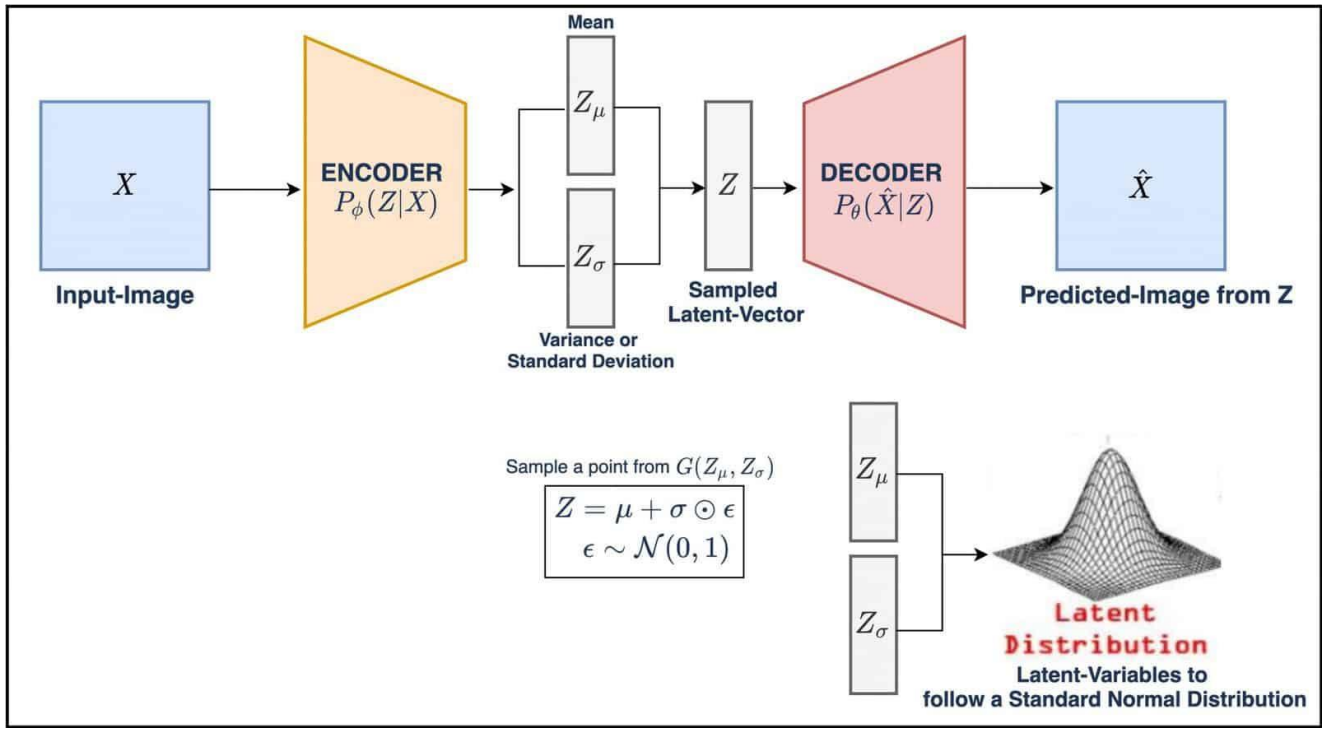
Original Deepfake

- One **shared encoder** + one **decoder per person**
- The shared latent keeps what's common: expression, pose, lighting
- **Identity lives in the decoder weights**
- At inference, swap the decoder → face A rendered as person B



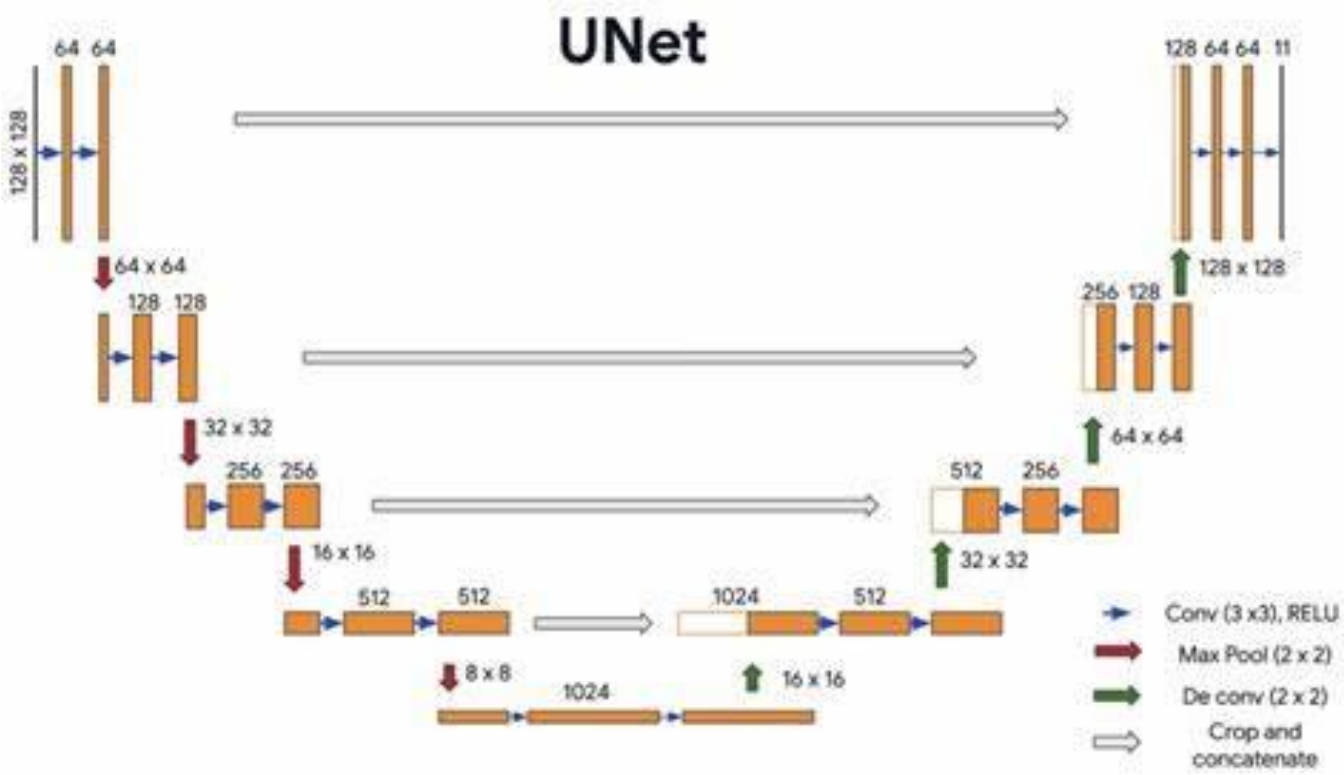
Architectures: Variational Autoencoders

- Plain AE: each input → a single point; gaps between points decode to garbage
- VAE: encode to a distribution → the latent space becomes smooth
- Smooth latent space → sample from it → generate new data
- Stable Diffusion runs on top of a VAE latent space



U-Net: A Relative, Not an Autoencoder

- Same encoder–decoder *shape* — but **not** an autoencoder
- Skip connections bypass the bottleneck
- Trained **supervised**: input → label (e.g., segmentation mask)
- Applications: medical image segmentation, diffusion model backbones



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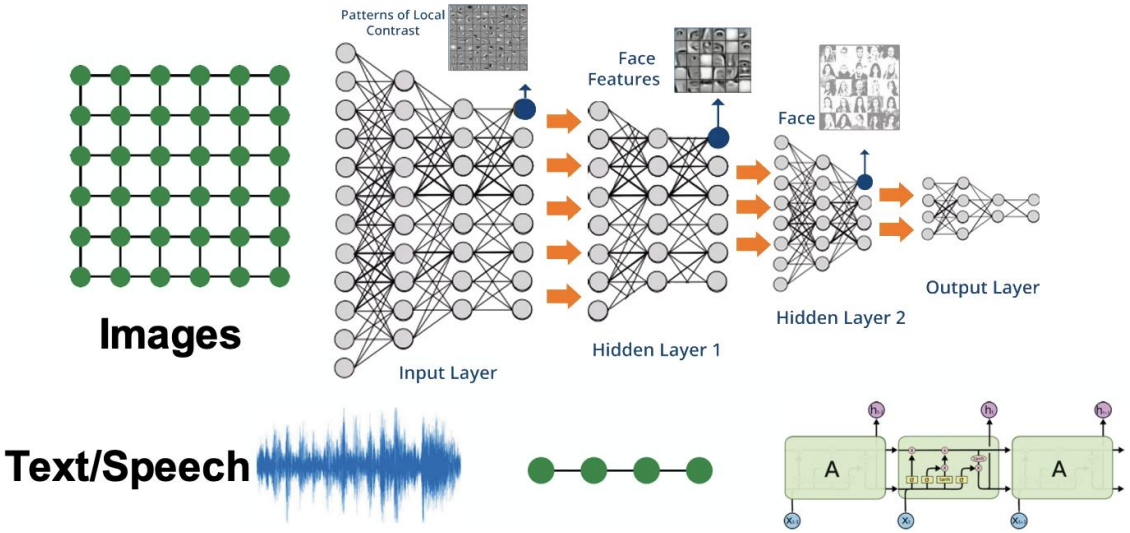
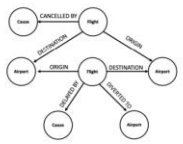
Graph Neural Networks

From Grids and Sequences to Graphs

- **Classical deep learning assumes regular structure.**
  - CNNs work on grids of pixels; RNNs on ordered tokens.
- **Most real data is irregular.**
  - Molecules, networks and maps are entities joined by relations.
  - There is no natural ordering, and degrees vary.
- **Graphs are the natural representation.**
  - GNNs extend convolution to arbitrary connectivity.



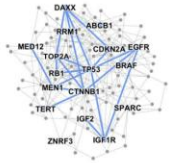
Graphs Are Everywhere



Event Graphs

Image credit: SalientNetworks

Computer Networks



Disease Pathways

Image credit: Medium

Social Networks

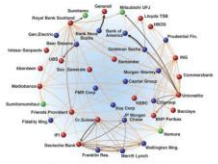


Image credit: Science

Economic Networks

Image credit: Lumen Learning

Communication Networks

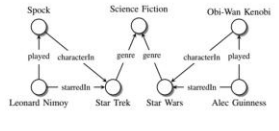


Image credit: Maximilian Nickel et al

Knowledge Graphs

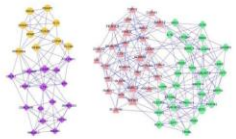


Image credit: ese.wustl.edu

Regulatory Networks

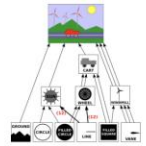


Image credit: math.hws.edu

Scene Graphs

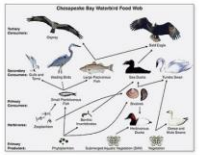


Image credit: Wikipedia

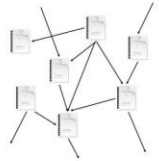
Food Webs

Image credit: Pinterest

Particle Networks

Image credit: visitlondon.com

Underground Networks



Citation Networks

Image credit: Missoula Current News

Internet

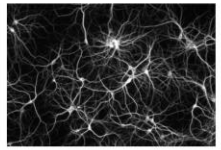


Image credit: The Conversation

Networks of Neurons

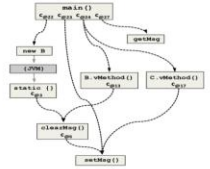


Image credit: ResearchGate

Code Graphs

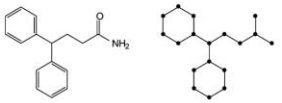


Image credit: MDPI

Molecules

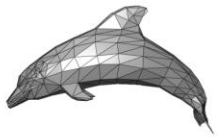


Image credit: Wikipedia

3D Shapes

Social and citation networks

Molecules and materials

Maps, meshes and 3D shapes

A Rapidly Growing Field

- GNNs went from niche to mainstream over the last decade.
- “Graph neural network” is now among the most common keywords at top ML venues.
- They are applied across chemistry, biology, recommendation, traffic and physics.

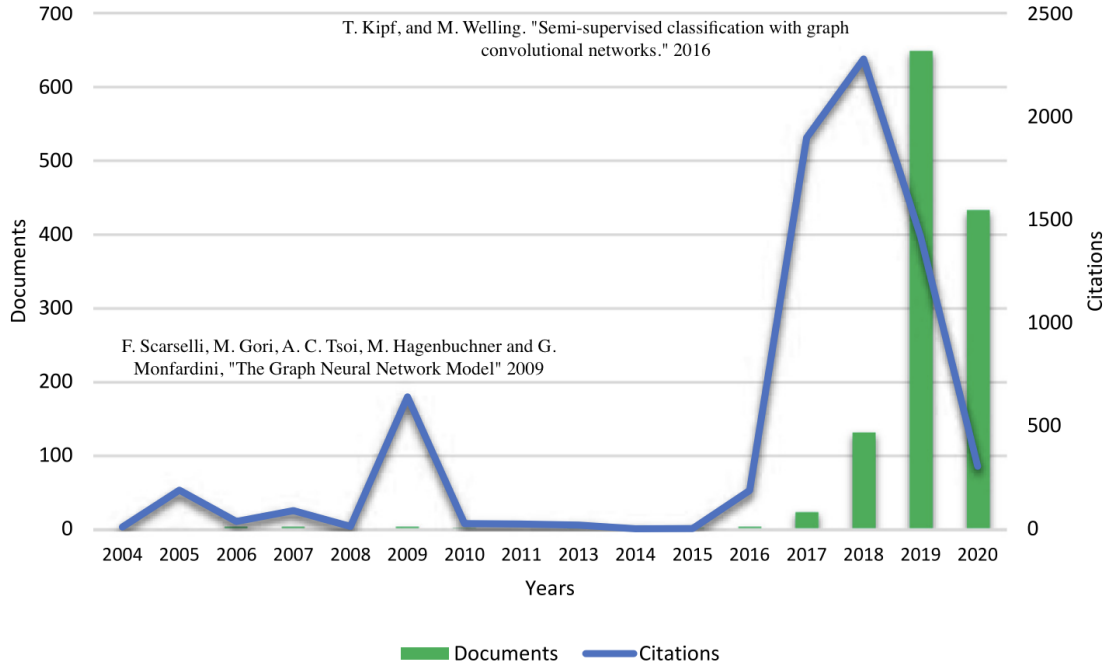
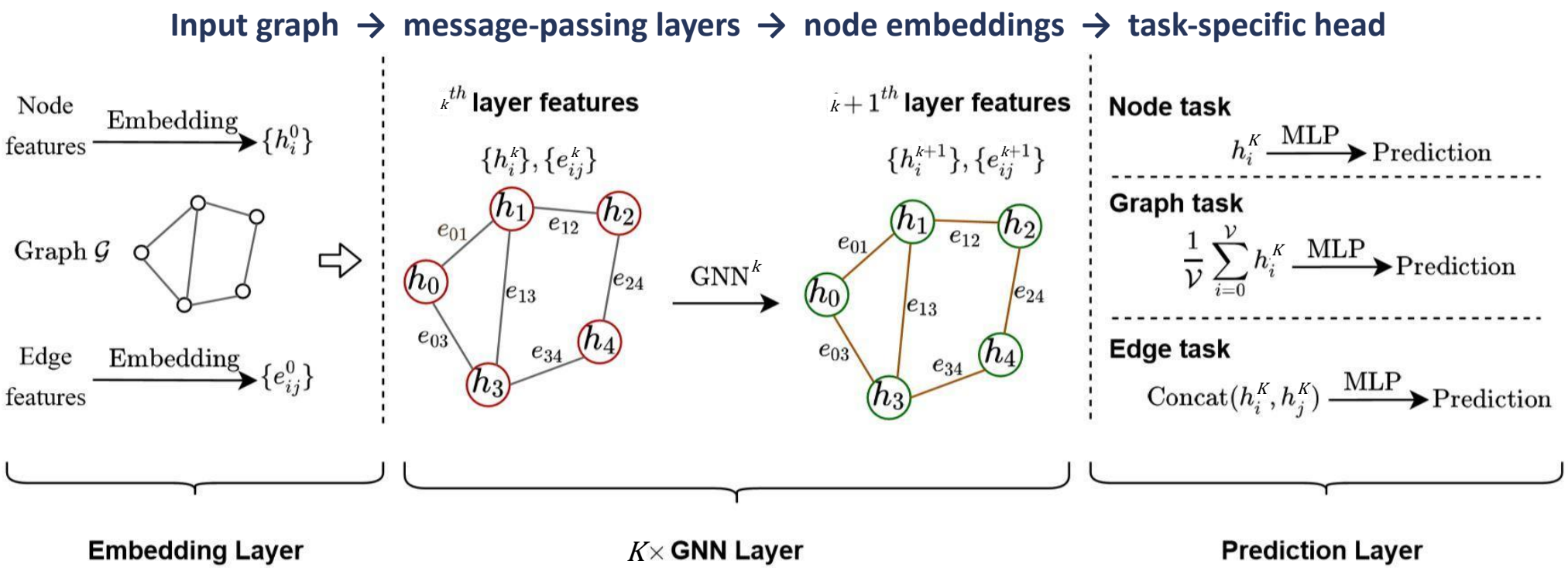


Fig. 1. Yearly number of documents and citations.

The GNN Pipeline



Node and edge level → Forecasting, routing, cybersecurity.

Graph level → Molecular properties, medical diagnosis.

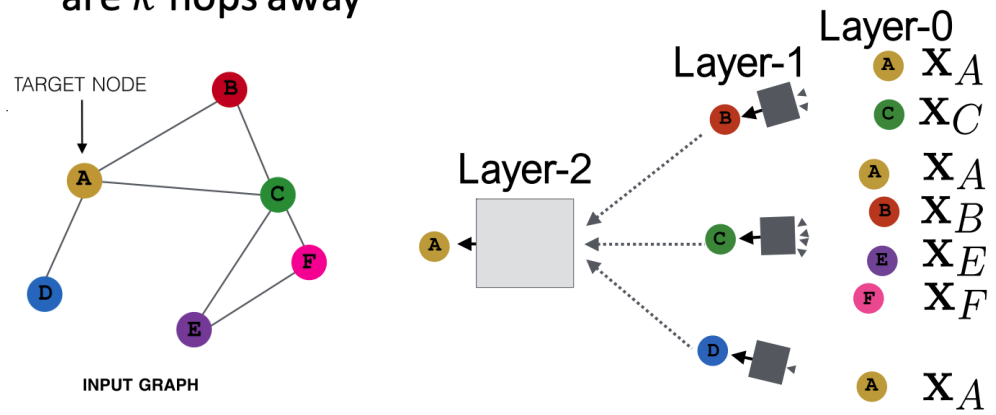
Link prediction → Recommendation, knowledge-graph completion.

Generation and counting → Molecule design, motif counting.

Message Passing

- **Each layer repeats two steps.**
  - Aggregate the neighbours' current embeddings.
  - Update the node's embedding with that aggregate.
- **Stacking layers grows the reach.**
  - After k layers a node has seen its k-hop neighbourhood.
- The result does not depend on node ordering.

- Model can be of arbitrary depth:
  - Nodes have embeddings at each layer
  - Layer-0 embedding of node v is its input feature, x_v
  - Layer-k embedding gets information from nodes that are k hops away

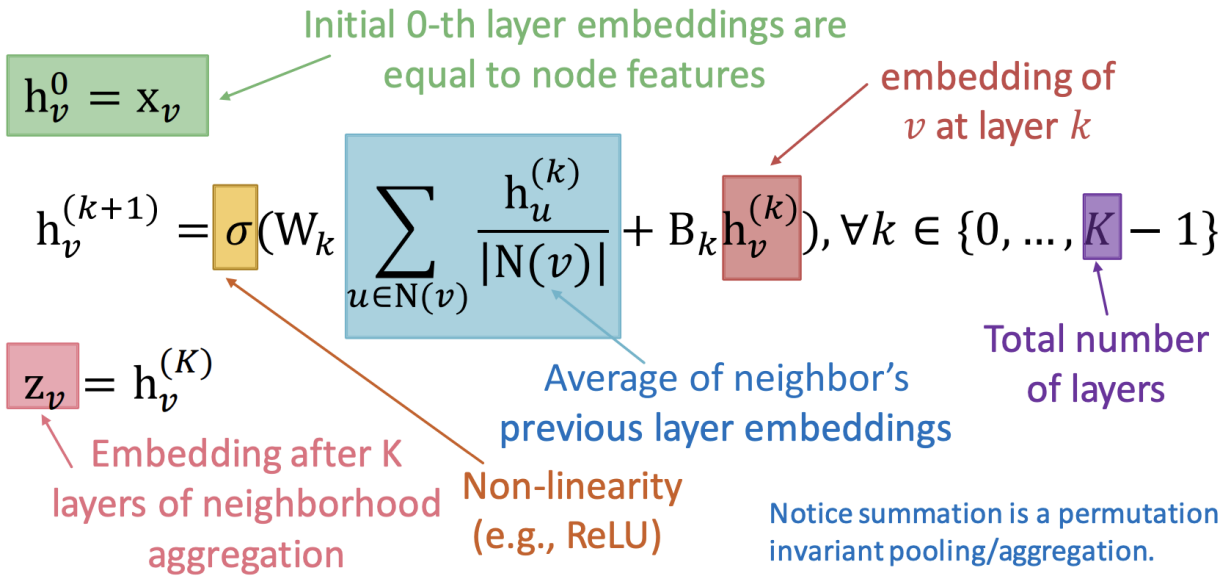


This is convolution, generalized — same weights, whoever your neighbours are.

The Graph Convolutional Network Update Rule (Canonical Modern GNN)

- Average the neighbours, then apply a learned weight and a nonlinearity.
- Symmetric normalisation keeps scales stable across node degrees.
- Weights are shared across the whole graph.
- Various aggregation choices → GAT(attention) / GraphSAGE(sampling) / GIN(sum)

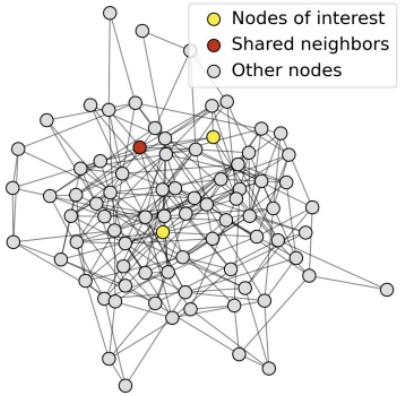
- **Basic approach:** Average neighbor messages and apply a neural network



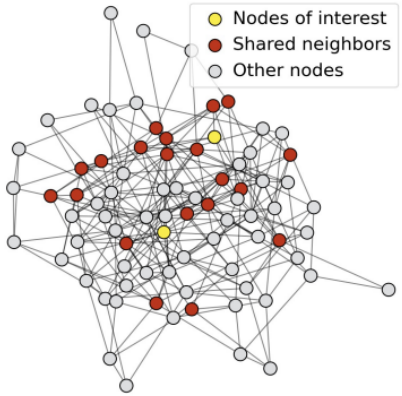
More Layers ≠ Better

- **Receptive field overlap** for two nodes
  - **The shared neighbors quickly grows** when we increase the number of hops (num of GNN layers)

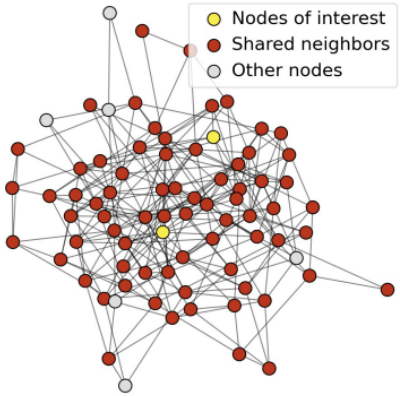
1-hop neighbor overlap
Only 1 node



2-hop neighbor overlap
About 20 nodes



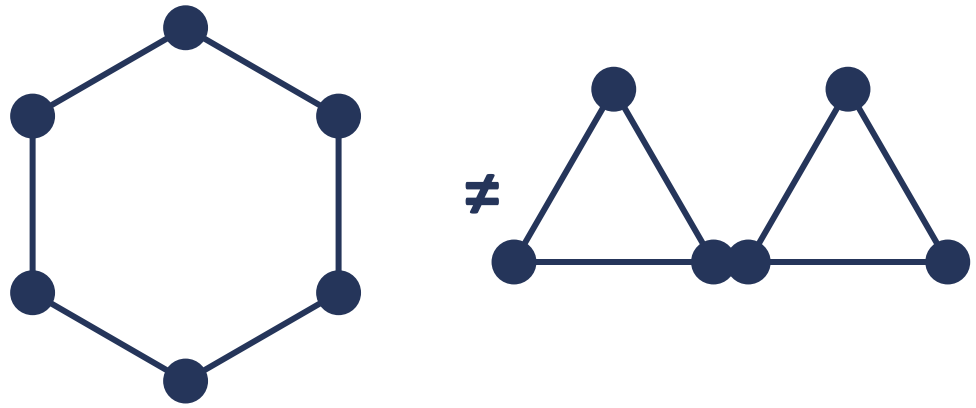
3-hop neighbor overlap
Almost all the nodes!



- **k layers = a k-hop field of view**
  - A few hops span most of a small-world graph.
- **Stack deeper and embeddings blur: oversmoothing.**
  - Fixes: residual connections, normalisation, jumping-knowledge.

Can a GNN Tell Two Graphs Apart?

- **A GNN describes each node by its neighbours,**
  - then its neighbours' neighbours, and so on.
- **If two graphs give the same descriptions,**
  - the GNN treats them as the same graph.
- So some genuinely different graphs are invisible to it.



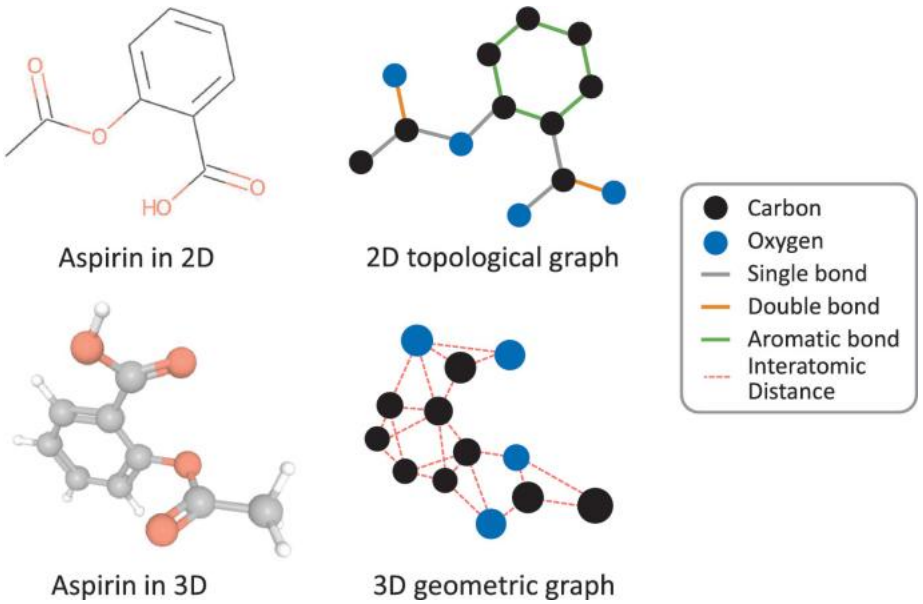
One ring of six **Two separate triangles**

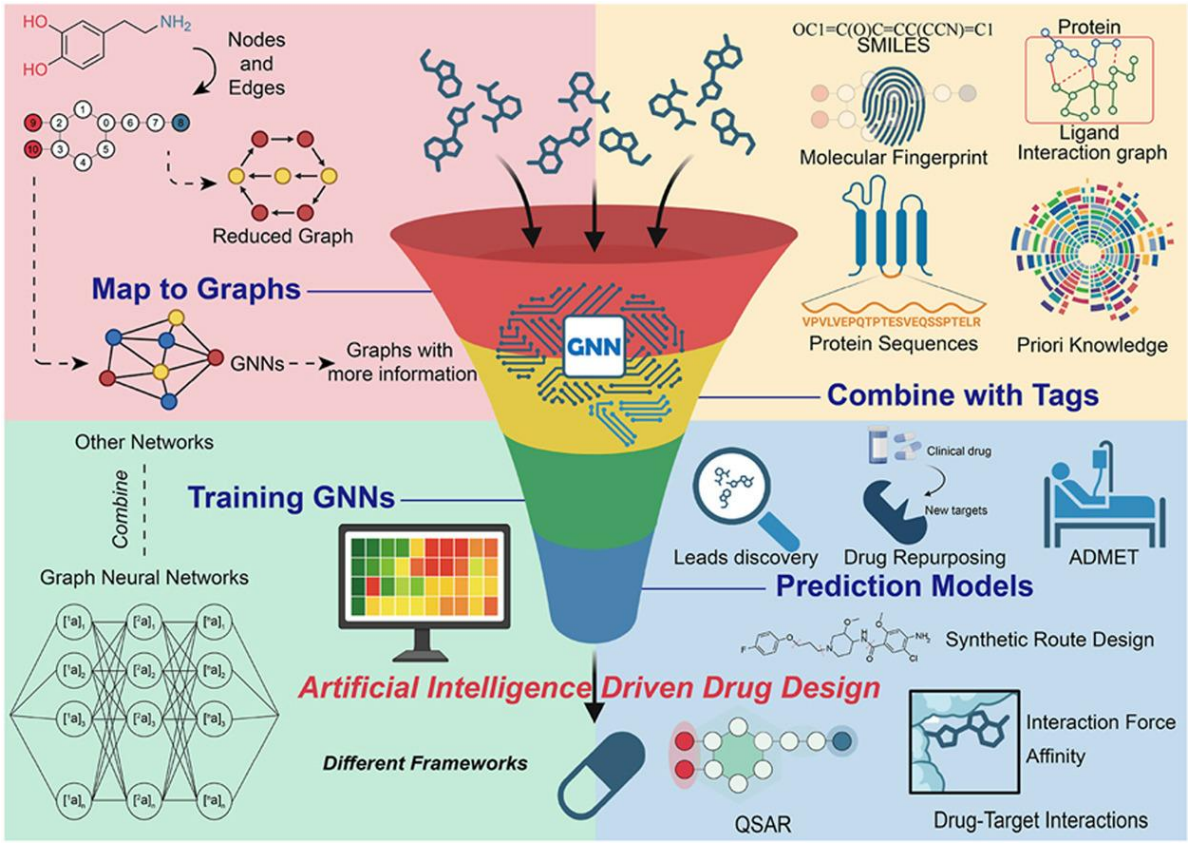
Every node has exactly two neighbours, so a standard GNN cannot tell these two apart.

Where GNNs Struggle

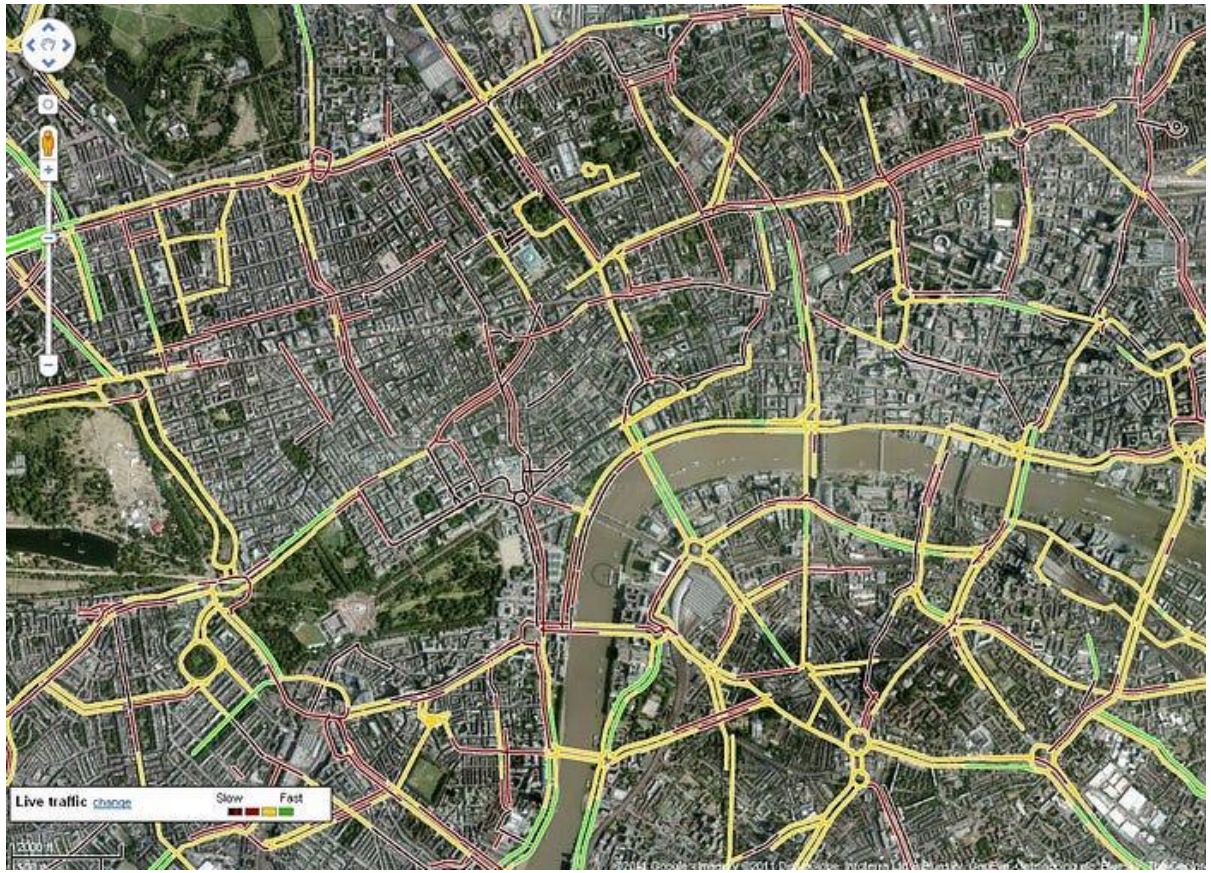
- **Heterophily**
  - Message passing assumes similar neighbours, and fails when they differ.
- **Oversmoothing**
  - Deep networks blur node embeddings together.
- **Oversquashing**
  - Long-range signal is crushed through bottlenecks.
- **Limited expressiveness**
  - Some genuinely different graphs look identical to a GNN.

Application – Molecules & Drug Discovery





Application – Google Maps ETA



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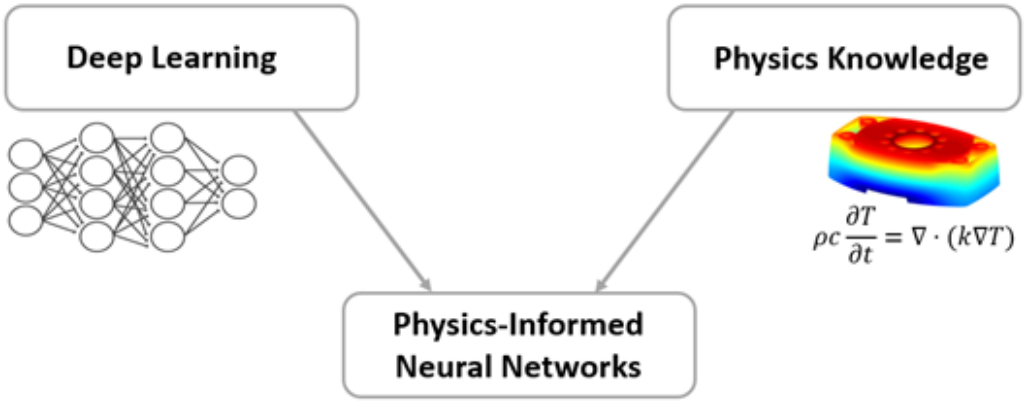
Physics-Informed Neural Networks

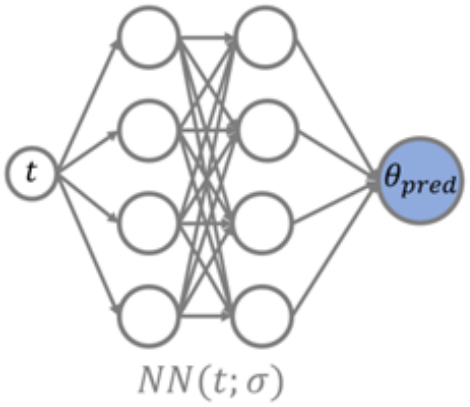
Why Physics-Informed Learning?

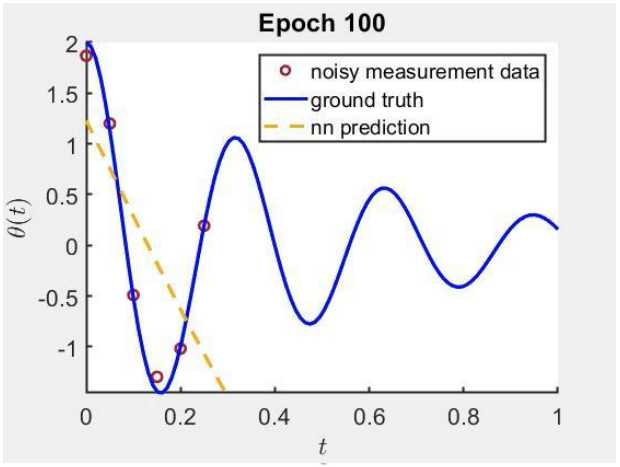
- **Standard networks fit data and ignore the physics behind it.**
- **Yet many systems obey known ODEs and PDEs.**
  - Fluids, heat, waves and structures all follow physical law.
- **PINNs add those laws to the loss.**
  - Better generalisation, less data, physically consistent (approximately) output.



Intuition: The Pendulum

Plain neural network

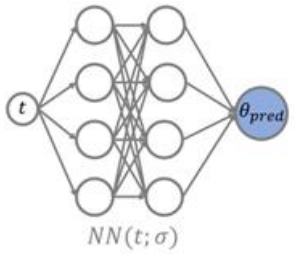




$$\min_{\sigma} \frac{1}{N} \sum_{i=1}^{N} \left| \theta_{pred}(t_i;\sigma) - \theta_{meas}(t_i) \right|^2$$

Fits the samples, but drifts from the true motion elsewhere.

Physics-informed network



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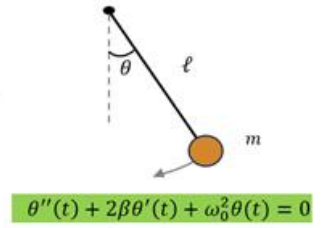
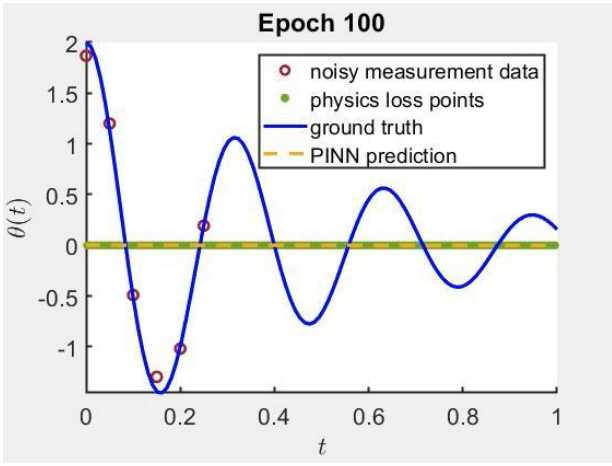


$$\theta''(t) + 2\beta\theta'(t) + \omega_0^2\theta(t) = 0$$



$$\min_{\sigma} \frac{1}{N} \sum_{i=1}^{N} \left| \theta_{pred}(t_i;\sigma) - \theta_{meas}(t_i) \right|^2$$

$$+ \frac{1}{M} \sum_{j=1}^{M} \left| \frac{\partial^2}{\partial t^2}\theta_{pred}(t_j;\sigma) + 2\beta \frac{\partial}{\partial t}\theta_{pred}(t_j;\sigma) + \omega_0^2\theta_{pred}(t_j;\sigma) \right|^2$$

The equation of motion is a loss term, so the fit stays physical even with sparse data.

https://www.mathworks.com/discovery/physics-informed-neural-networks.html

From a Niche Idea to Scientific ML

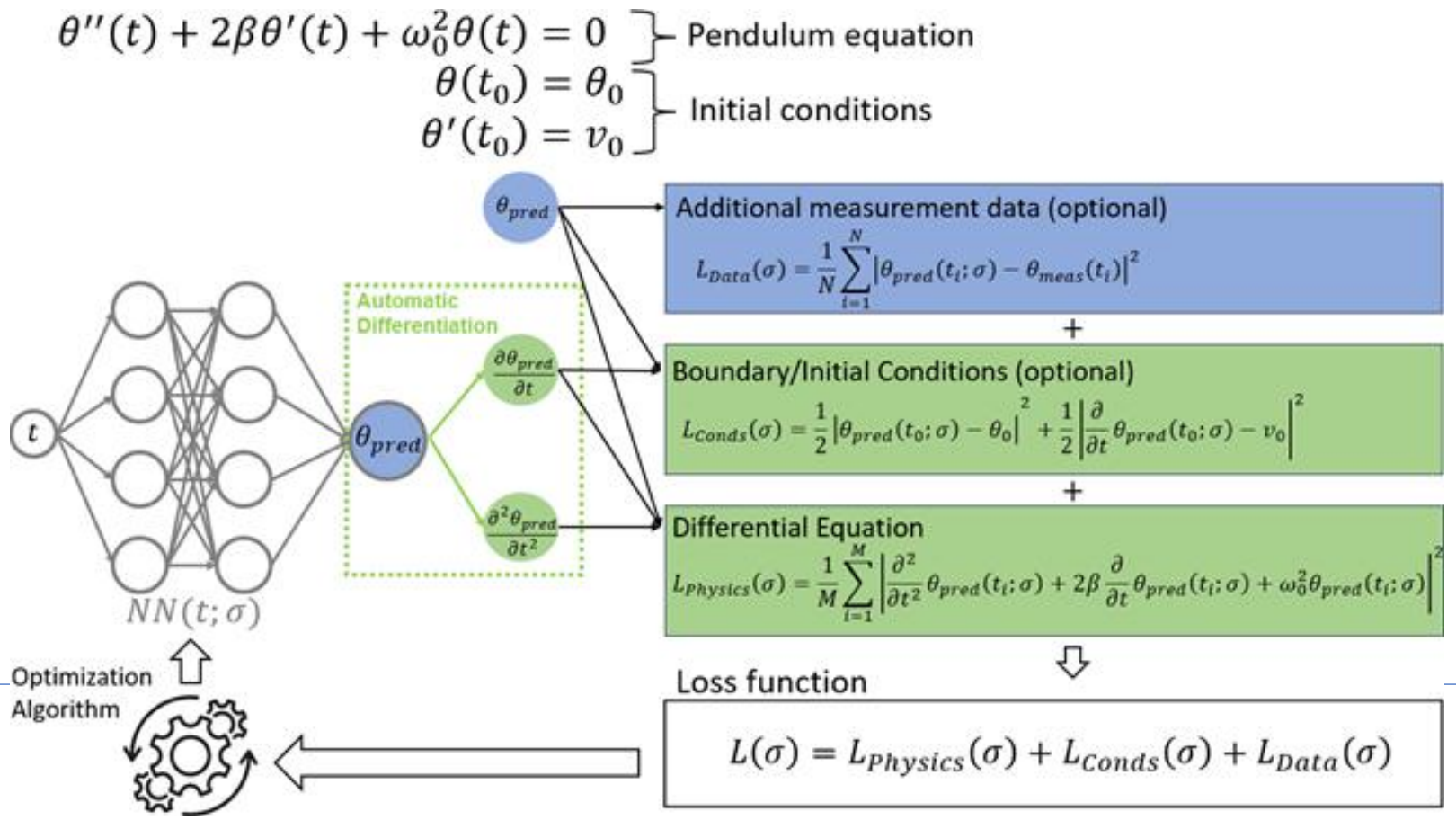
- **1998 — First steps**
  - Lagaris et al. solve differential equations with neural nets, limited by the compute of the day.
- **2017–19 — Modern revival**
  - Raissi, Perdikaris and Karniadakis reframe and popularise PINNs.
- **Today — Scientific ML**
  - A standard tool for simulating and inferring physical systems.
- **What made PINNs possible:** automatic differentiation, powerful GPUs, and mature optimizers such as Adam.

Where PINNs Shine

- **Expensive simulations**
  - Replace costly FEM or CFD solvers with fast approximations (surrogate modelling).
- **Sparse or noisy data**
  - Use physics to fill gaps, as in flow reconstruction or seismology.
- **Inverse problems**
  - Infer parameters that cannot be measured directly.

Ingredients

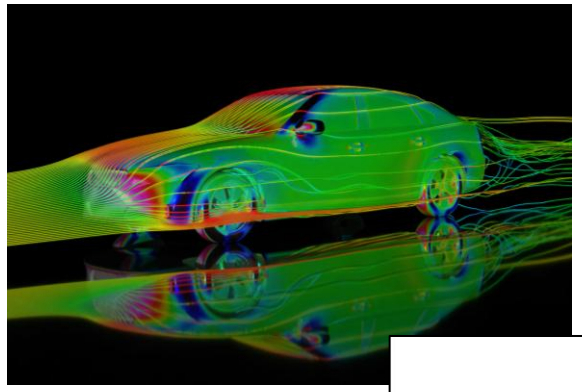
- Derivatives come straight from the computational graph (Autodiff).
- The objective: $L_{total} = L_{data} + L_{IC/BC} + L_{phys}$

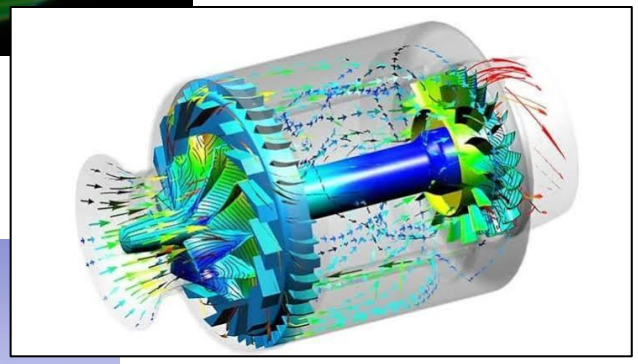


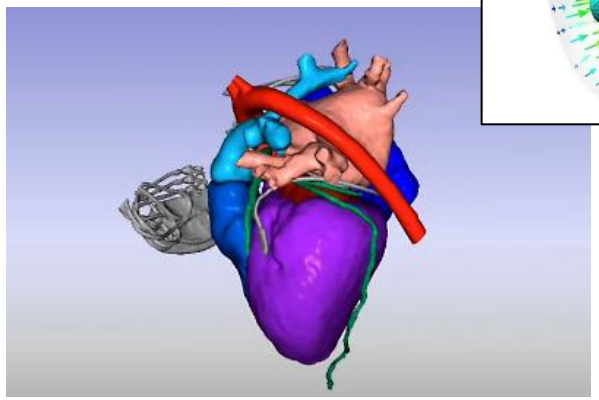
Forward and Inverse Modeling

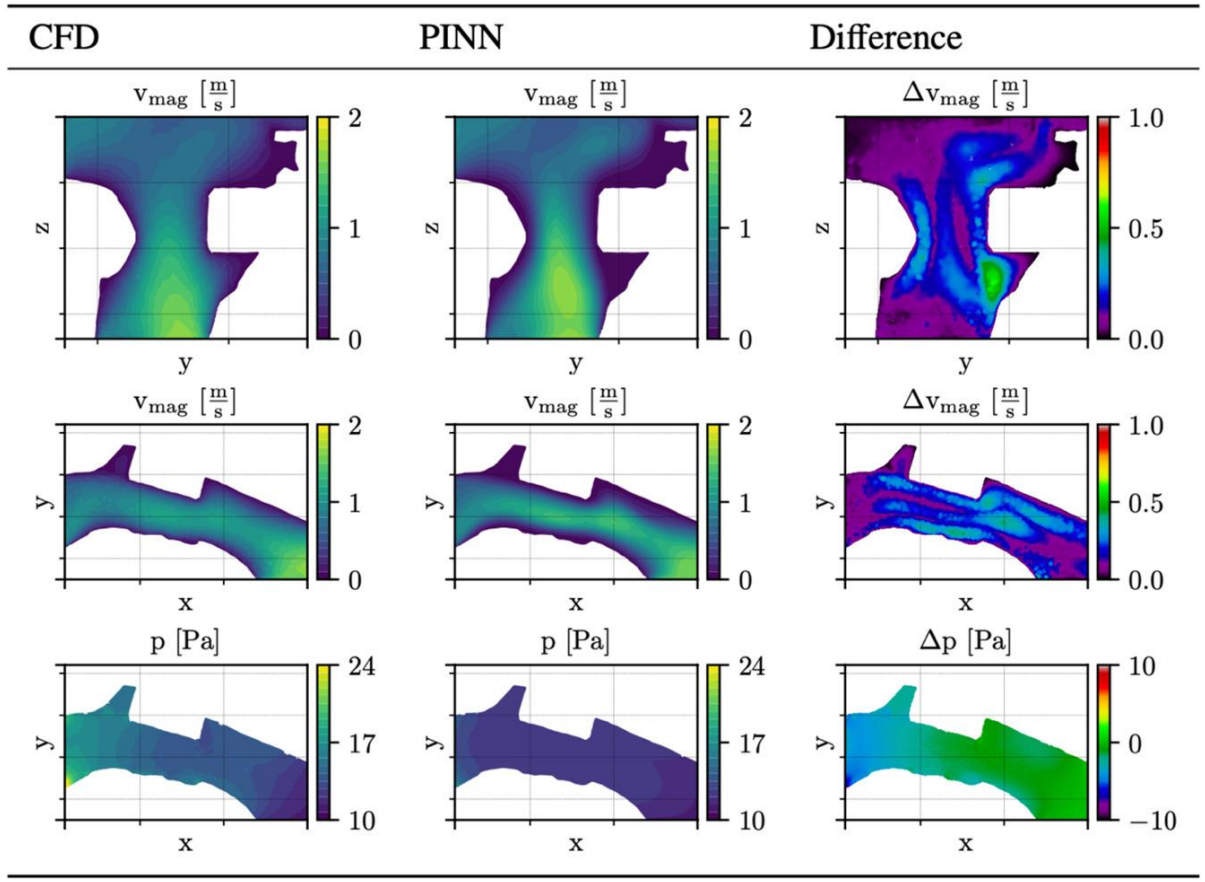
- **Forward:** given parameters, compute the solution field.
  - e.g. given conductivity K, solve for the temperature.
  - Replaces an expensive simulator.
- **Inverse:** given sparse observations, infer hidden parameters.
  - e.g. estimate K from measured temperatures.
  - Ill-posed: sensitive to noise and often non-unique.
- Trainable quantities can include not just the weights, but unknown physical parameters.

CFD Surrogate Modelling









The Hard Parts

- **Training difficulty**
  - Stiff, multi-scale dynamics and heavy problem-specific tuning.
- **Computational cost**
  - Residuals and high-order derivatives are expensive to evaluate.
- **Generalisation**
  - A trained PINN usually fits one setup and must be retrained.
- **Inverse modeling**
  - Ill-posed and non-unique; several parameters need more data.

The Same Idea in Other Domains

- **Physics**
  - Conservation laws and PDEs must hold.
- **Finance**
  - Prices must be arbitrage-free.
- **Combinatorial optimization**
  - Solutions must stay feasible.
- **When to reach for a PINN:** the rules matter more than raw accuracy, and labelled data is scarce.

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Thank you.